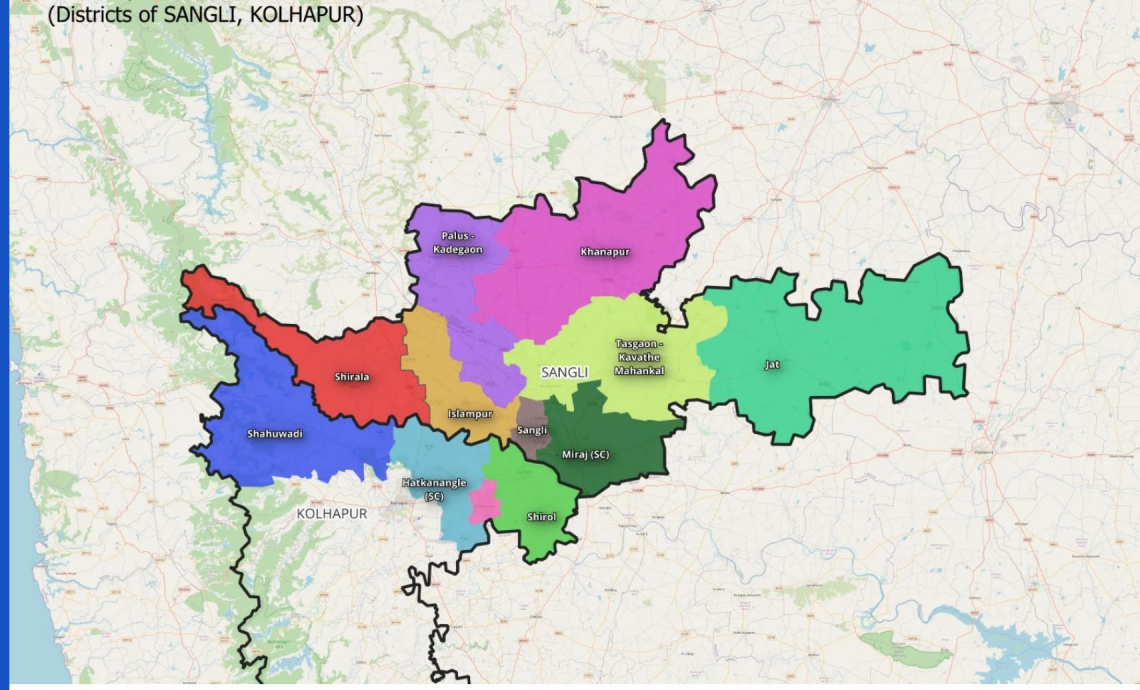
IE301 OPERATIONS ANALYSIS
GROUP 4, PRESENTATION 3

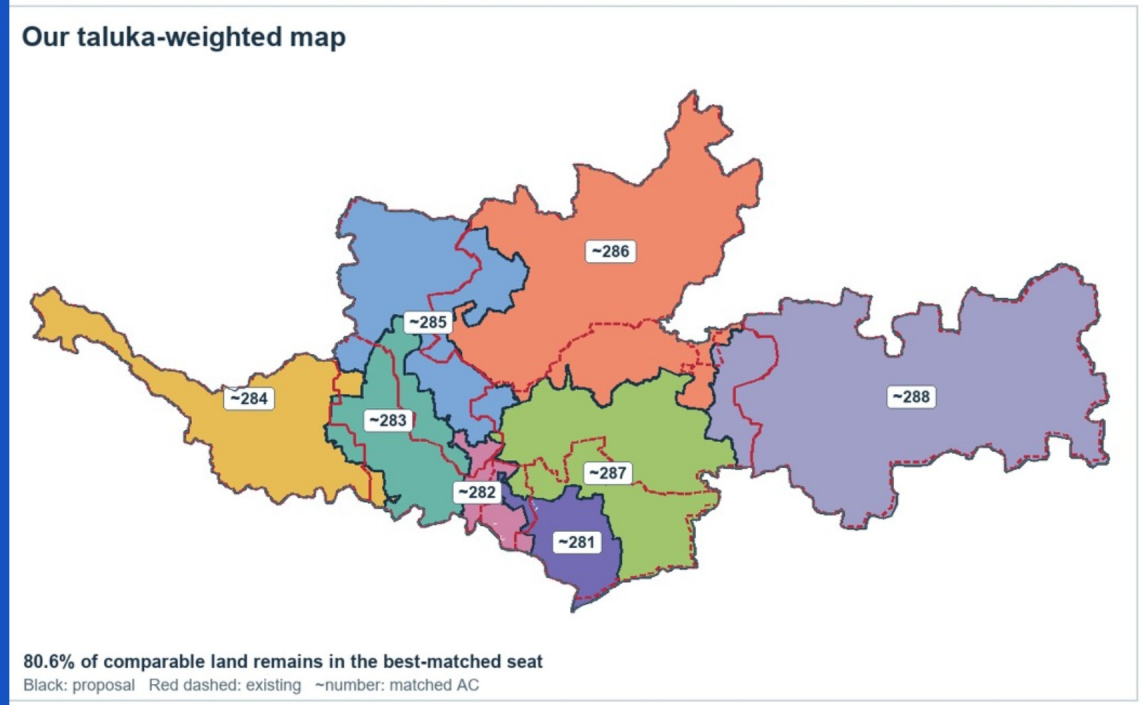
Election Logistics

Districting Sangli into eight constituencies, then placing polling booths inside each

Pari Soni (24b4505), Shivam Soni (24b4525)
Shreyaan Sharma (24b4529)
Asmaniwala Devanshu (26x0053), S Ganesh (26x0052)

Mentor: Prof. Ketan Savla





Recap of Presentations 1 and 2

1 Constituency design

Where do the boundaries go?

Population | Geography | Demographics

Fair representation: each representative serves roughly the same number of people.

2 Operational delivery

Can people actually vote?

Booth capacity | Budget | Travel time

Real access: compact, connected seats keep the trip to the booth manageable.

Same voters → **Different boundaries** → **Different results** *= gerrymandering risk*

DATA STRUCTURE

3 sources
Census 2011, SHRUG, city wards

→

756-row unit table
taluka, people, point, polygon

→

Border graph
756 nodes, 1,988 edges

→

Decision variable
seat(i) ∈ {1, …, 8}

728 villages | **6** towns | **20** city wards | **2** empty land gaps | **10** talukas | median village **1,927** people | **5.3** neighbours per unit

Problem Setup: Sangli District

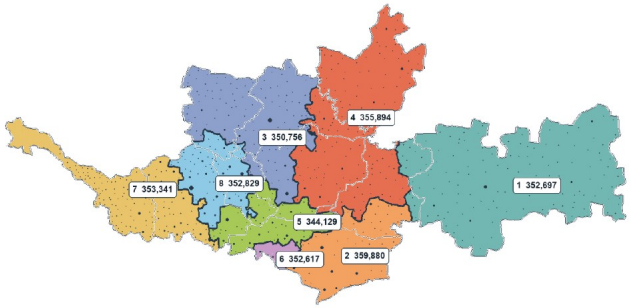
2.82 M
people, Census 2011

8
Assembly seats

756
settlement units

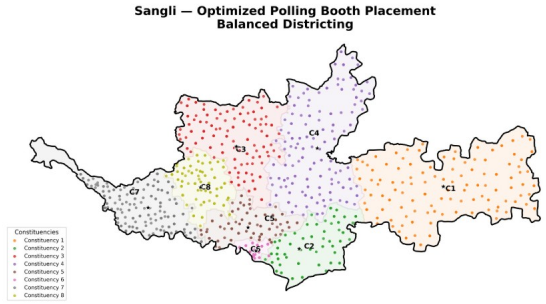
352,768
target per seat, ±10%

Stage 1: Draw 8 Seats



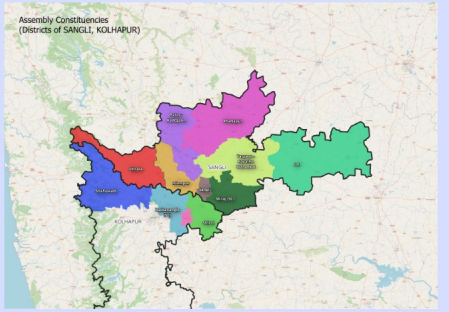
Each seat connected,
317,492 to 388,044 people

Stage 2: Place Booths



Seats fixed,
≤ 1,500 voters per booth

Stage 3: Choose a Map



Compare with today's seats
on score and voter travel

Methods Explored

WHAT MAKES A GOOD MAP — **A** Population 30% · **B** Contiguity Hard · **C** Compactness 20% · **D** Logistics 15%

Genetic algorithm

Evolve a population of candidate maps

- Start from random connected maps
- Score on balance, shape, logistics
- Select, cross over, mutate, repeat

✓ Explores a huge space

⚠ Crossover breaks contiguity

Quadtree

Split into 4 until each cell is under a population threshold

- Simple recursive procedure
- Merge small cells back up
- Fast and easy to explain

✓ Quick first cut

⚠ Ignores shape and contiguity

Graph partitioning

Villages are nodes, borders are edges; split into k balanced groups

- Kernighan-Lin: local swaps
- Multilevel: coarsen, split, refine
- Weighted k-means: penalise heavy clusters

✓ Natural adjacency model

⚠ Connectivity must be enforced

ReCom — Adopted

Merge two neighbouring seats, redraw a spanning tree, re-split

- Seats stay connected; villages stay whole
- Hard rules: ±10% population, no thin necks
- Four-factor score with annealing

✓ Valid map at every step

⚠ Heuristic: no optimality proof

WHY RECOM — ✓ Seats connected by construction · ✓ Whole villages never cut · ✓ Samples many legal maps, keeps the best

First Attempt: Genetic Algorithm

1 **Divide into 8 seats**
727 Census units

2 **Place booths per seat**
solved together with 1

ONE EVALUATION OF A CANDIDATE MAP

GA proposes 8 seats → **Booths optimised** → **Costs computed** → **Map scored J**

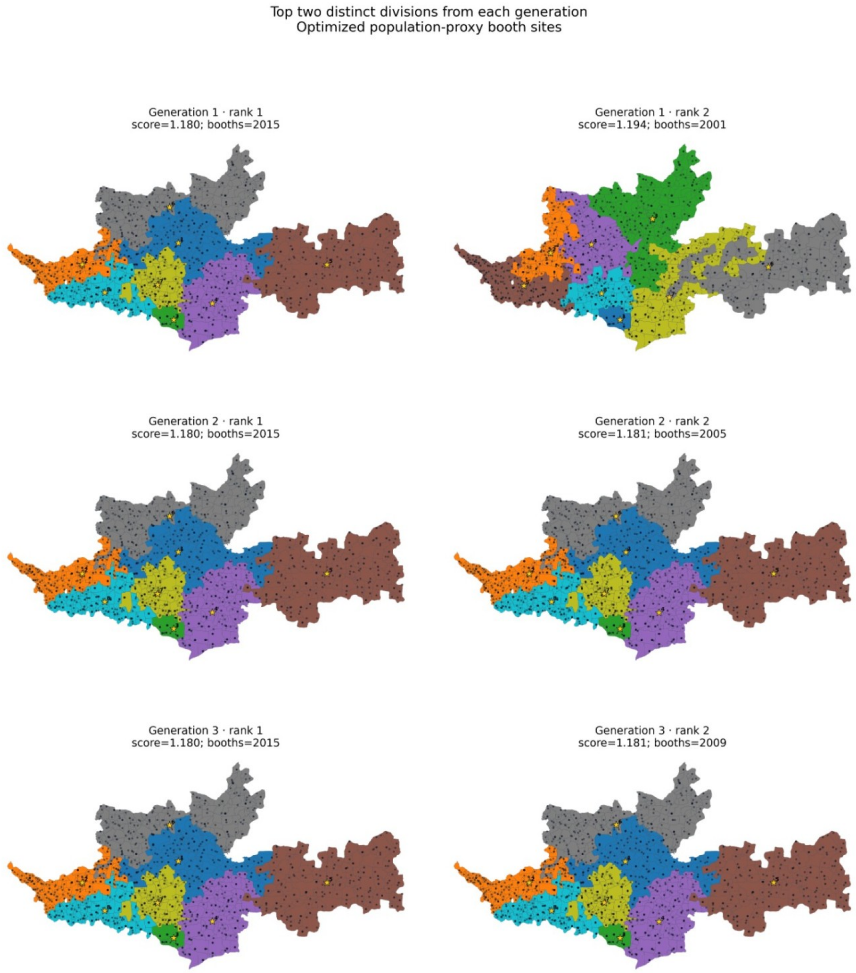
Best maps survive, cross over and mutate; the loop repeats each generation.

$$J = 0.30\,\mathrm{Pop} + 0.20\,\mathrm{Shape} + 0.35\,\mathrm{Access} + 0.15\,\mathrm{Logistics}$$

Lower J is better. Contiguity is a hard rule, outside J.

WHAT THE RUNS SHOWED

- ⚠ **Stalled early:** best J stays at 1.180 from generation 1 to 3
- ⚠ **Messy runners-up:** stringy, spread-out seats after crossover
- 🕒 **Slow to score:** booth placement re-runs for every candidate
- 💡 **So we switched districting to ReCom and moved booths to a second stage**



Districting Model: ReCom

MUST PASS $317{,}492 \le P_s \le 388{,}044$ | Connected | No neck under 0.5 km

$$\text{Score} = w_B\, B + w_D\, L_D + w_F\, L_F + w_C\, C$$

Balanced 35/30/20/15 | **Locality** 30/45/10/15 | **Taluka** 30/20/35/15

B: population balance

$$B = \frac{1}{K}\sum_s \frac{|P_s/T - 1|}{0.10}$$

D: distance to seat centre

$$L_D = \frac{1}{20\,\text{km}} \cdot \frac{\sum_s P_s D_s}{P}$$

F: taluka fragments

$$L_F = \frac{1}{8}\sum_t \max(0,\ q_t - m_t)$$

C: shape compactness

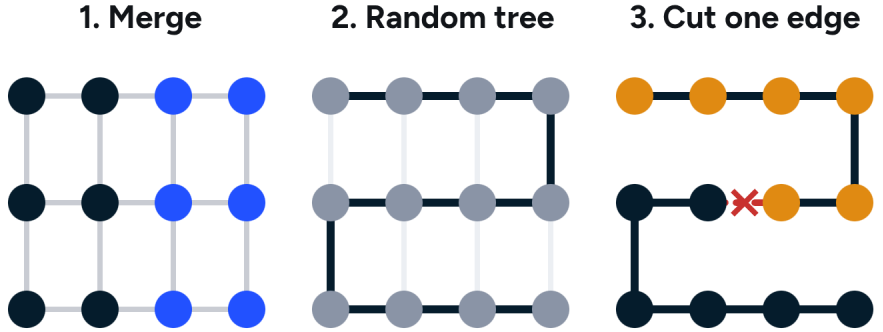
$$C = \frac{1}{K}\sum_s \left(1 - \frac{4\pi A_s}{\text{perim}(G_s)^2}\right)$$

Simulated annealing: accept some worse maps early, fewer later

$$P(\text{accept}) = \min\left(1,\ e^{-\Delta/T}\right), \quad T: 0.05 \to 0.01$$

Δ = new score − current score. Lower score is better; the best valid map is kept.

One ReCom move, on 12 toy units



Merge: two neighbouring seats (6 + 6 units) become one region.

Tree: link all 12 units with no loops, using real borders only.

Cut: remove the one edge that leaves two halves of equal population. Both stay connected.

Search effort	Proposed	Accepted	Rate
Balanced	600	314	52%
Locality	600	378	63%
Taluka	300	165	55%

Polling Booth Model

1 Size each seat → 2 Spread seeds → 3 Assign ≤ 1,500 → 4 Move to centroid

Repeat 3 and 4 until stable

Booths needed per seat

$$B_j = \left\lceil \frac{P_j}{1500} \right\rceil$$

Voter travel Minimised

$$C_V = \sum_i \sum_b P_{i,b}\, d(i,b)$$

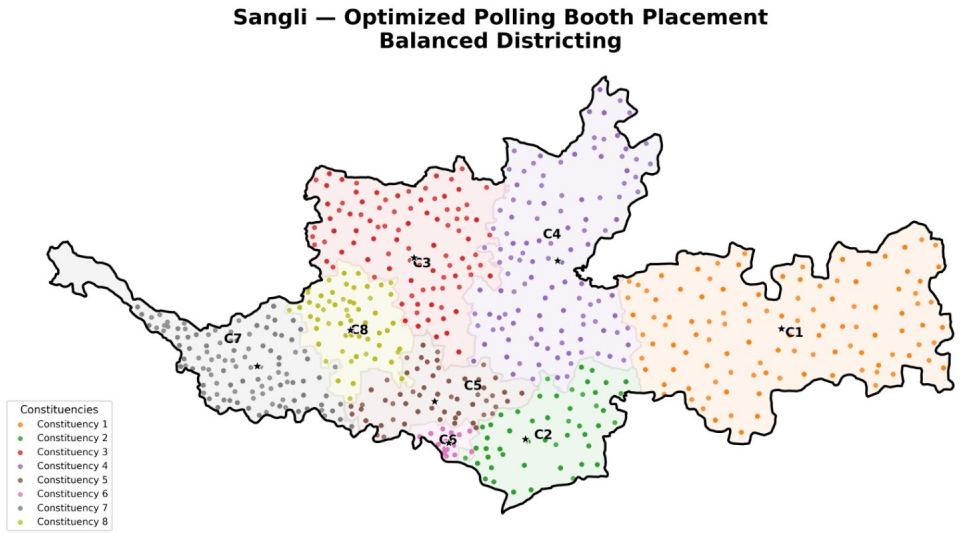
Booth moves to weighted centroid

$$x_b^* = \frac{\sum_i P_{i,b}\, x_i}{\sum_i P_{i,b}}$$

EVM transport Reported

$$C_E = \sum_b d(W_j,\ b)$$

Straight-line km $d(i,b) = \sqrt{(x_i - x_b)^2 + (y_i - y_b)^2}$

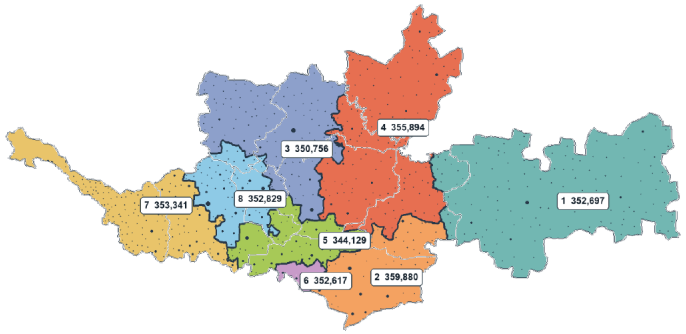


Balanced map: dots are booth sites, stars are seat centres

Districting Results

Map A: Balanced

35/30/20/15



2.45%
max pop. deviation

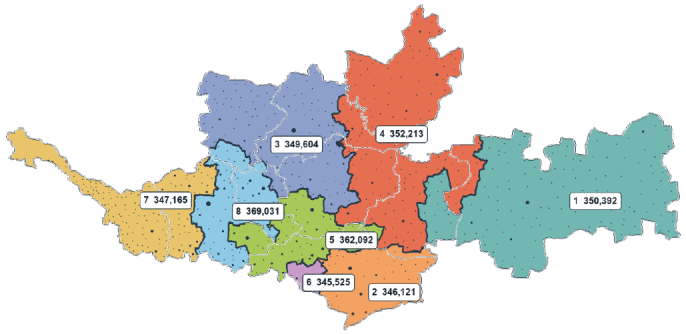
12
extra taluka pieces

0.629
common score

67.9%
land kept vs today

Map B: Locality

30/45/10/15



4.61%
max pop. deviation

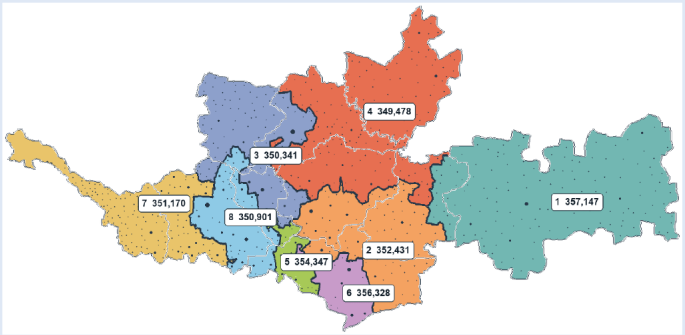
13
extra taluka pieces

0.687
common score

66.6%
land kept vs today

Map C: Taluka

Best on 4 of 4



1.24%
max pop. deviation

11
extra taluka pieces

0.602
common score

80.6%
land kept vs today

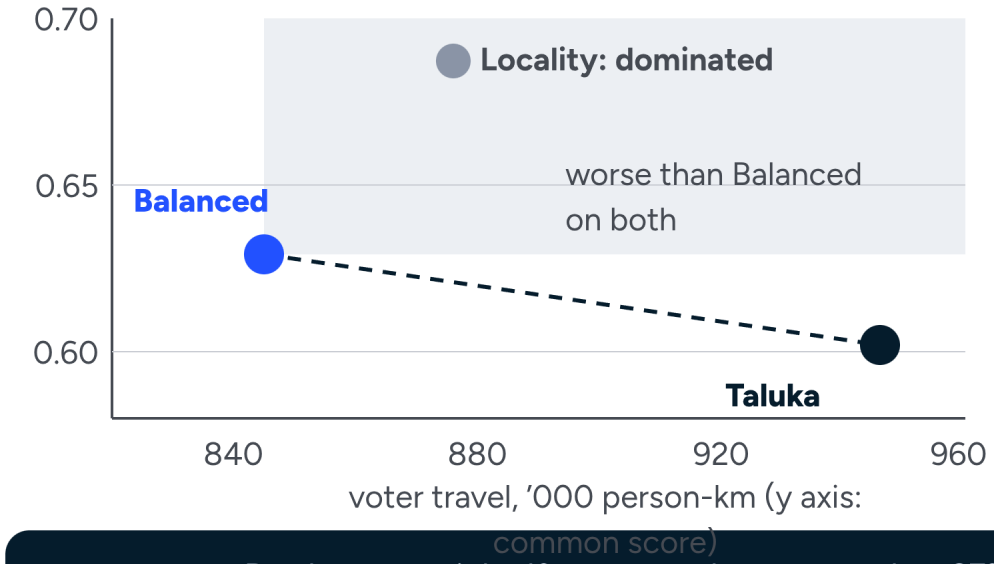
⚠ Average distance to the seat centre is about 12.7 km in all three maps, so it does not separate them. Within a map it runs from 3 km (city seat) to 21 km (Jat seat).

Logistics Results and Decision

Criterion (lower is better)	Balanced	Locality	Taluka
Max population deviation	2.45%	4.61%	**1.24%**
Extra taluka pieces	12	13	**11**
Common districting score	0.629	0.687	**0.602**
Voter travel ('000 person-km)	**845**	876	946
Avg. voter distance (km)	**0.299**	0.310	0.335
EVM transport ('000 km)	23.92	**23.78**	23.78
Booths needed (min. 1,882)	1,886	1,886	**1,885**
Land kept vs today (higher is better)	67.9%	66.6%	**80.6%**
Pareto status (score vs travel)	**Efficient**	**Dominated**	**Efficient**

Bold blue marks the best value in each row. EVM transport and booth counts differ by under 1%, so they do not decide.

Trade-off: districting score vs voter travel



27% Break-even weight. If voter travel gets more than 27% of the weight, Balanced beats Taluka: 4.5% score gap vs 12.0% travel gap.

Future Scope

1 Better distances

Now Straight-line km from one point per village

Next
- Road-network shortest paths (OpenStreetMap)
- Travel time, not km: terrain, rivers, ghats
- Spread people inside each village



Dashed: today **Solid: road path**

2 One joint model

Now Draw seats first, check logistics after

Next
- Booth and EVM cost inside the ReCom score
- Fast booth estimate per proposal, full solve on finalists
- Weights set from the Pareto trade-off

$$\text{Score} + w_V \frac{C_V}{C_V^{\min}} + w_E \frac{C_E}{C_E^{\min}}$$

one objective for both stages

3 Scale to Maharashtra

Now One district pilot, 756 units

Next
- Run each district in parallel
- Seats stay inside district lines
- Same rules, one state-wide report

288
seats (Sangli: 8)

112 M
people (Sangli: 2.8 M)

Feasible booth sites
schools, offices, polling stations

District EVM flow
district → seat warehouse → booth

Real-world limits
terrain, security, temporary booths

Electors, not population
plus a baseline on today's map